

Peer-Isolated Recommendation: When Content-Only Personalization Beats Collaborative Signal, and When It Doesn't

Bidit Das

Independent Researcher | Dallas, TX

biditdas18@gmail.com

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Abstract

We present **Essence**, a personal embedding cluster framework for recommendation that operates exclusively within a single user's engagement history, without consulting cross-user interaction data. We evaluate Essence on three datasets spanning distinct domains — Last.fm-1K (music), Amazon Books (e-commerce), and MovieLens-25M (film) — against a real collaborative-filtering baseline, a non-personalized popularity baseline, a content-based average-embedding baseline, three recency-based baselines (last-item, uniform-average, and exponentially-weighted retrieval), and two hand-implemented multi-interest neural baselines (MIND [1], ComiRec [2]).

Essence's performance relative to these baselines is **domain-dependent**: across the three evaluated datasets, Essence's rank among the ten evaluated systems is associated with the strength of collaborative/popularity signal available in the domain; establishing this as predictive would require additional datasets and an independently-defined domain property, not one measured post hoc via baseline performance. K (the number of interest clusters) is selected per dataset on a validation split carved from training data, not fixed in advance — $K = 10$ for Last.fm-1K and Amazon Books, $K = 15$ for MovieLens-25M. On Amazon Books, where collaborative and popularity baselines are structurally weak (Recall@10 of 0.0031 and 0.0021 respectively), Essence ranks 3rd of 10 on Recall@10 (0.0253): it loses only to Last-Item (0.0311) and is statistically tied with, not significantly behind, Recency-Weighted (0.0260, $d = -0.009$) and Avg-Last-10 (0.0235, which it now numerically beats). It significantly beats collaborative filtering, popularity, content-averaging, and both neural multi-interest baselines on Recall@10 (d ranges 0.099–0.282), and significantly beats Random, Popularity, and MIND on LT-Recall@10 specifically; it is statistically tied with CF-ItemKNN, Content, ComiRec, Last-Item, Recency-Weighted, and Avg-Last-10 on LT-Recall@10. On Last.fm-1K, Essence ranks 2nd of 10 on both Recall@10 (0.0281) and LT-Recall@10 (0.0142); it is not significantly different from any of the three recency-only baselines (numerically ahead of Last-Item and Avg-Last-10, behind Recency-Weighted, 0.0304, $d = -0.127$, not significant). On MovieLens-25M, where collaborative signal is strong (item-based CF reaches 0.0850 Recall@10, driven almost entirely by the platform's densely-rated popular titles), Essence ranks 8th of 10 systems, losing to collaborative filtering, popularity, and both neural baselines by the largest effect sizes observed in this study (Cohen's d down to -0.55). A silhouette-stratification test, comparing Essence against Recency-Weighted within per-user clustering-quality tertiles, finds that K -means clustering is not empirically supported as the source of Essence's advantage where an advantage exists: no stratum, on any dataset or metric, shows Essence significantly ahead of Recency-Weighted, though only one of sixteen clean such comparisons (MovieLens-25M's highest-silhouette stratum on LT-Recall@10) now shows Essence significantly *behind* it either, down from six of seventeen before validation-selecting K . The question this paper does not fully resolve — how much of Essence's advantage, where it exists, comes from clustering specifically versus recency-weighted retrieval alone — remains open.

We report every comparison with paired bootstrap significance testing (10,000 resamples), Benjamini–Hochberg correction for multiple comparisons, standardized effect sizes, and BCa robustness cross-checks — replacing an earlier confidence-interval-overlap methodology

that does not test the quantity of interest. We also report a popularity-decile breakdown showing that Essence’s long-tail advantage, where present, is concentrated in moderately-rare-but-observed items rather than true cold-start items, and an artist/author-identity shortcut analysis showing that a disproportionate share of Essence’s hits come from same-artist/author candidates despite these being a minority of its recommendations. We frame Essence not as a universally superior candidate generator, but as a data point on when peer-isolated, content-only personalization is and isn’t the right architectural choice. Code: <https://github.com/biditdas18/essence>

1 Introduction

Recommendation systems built on collaborative filtering (CF) derive their signal from cross-user interaction patterns: a user’s future preferences are predicted from the behavior of users who interacted similarly in the past. This approach performs well on standard benchmarks, particularly on metrics dominated by popular items, but it carries a structural property worth examining directly: items with sparse interaction histories are, by construction, underrepresented in any signal derived from cross-user co-occurrence. A user with a narrow, idiosyncratic taste profile (engaging consistently with low-interaction items across a domain) receives recommendations shaped by aggregate similarity to other users, not by the specific combination of properties their own history reflects.

This work originated from a practical observation in music recommendation: users with precise, narrow taste profiles are reliably steered toward broadly popular artists that are only loosely similar, while artists matching the specific texture of that taste remain unsurfaced regardless of how long the user searches the catalog manually. This pattern motivated two questions addressed here. This paper is structured as an evaluation study, not a method-introduction paper: the central question is whether a specific architectural choice — K -means interest clustering — provides measurable benefit over a simpler recency-based alternative, tested under conditions designed to give that choice its best chance to matter. First, the original motivating question: can a recommendation signal be constructed using only a single user’s own history, without any cross-user data, and if so, does it recover items that collaborative methods structurally underrepresent? Second, a question that only becomes answerable once the same comparison is run across multiple, structurally different domains: under what domain conditions does this approach help versus hurt, and is the effect predictable from properties of the domain rather than discovered post hoc per dataset? A one- or two-dataset evaluation cannot distinguish a domain-general advantage from a domain-specific artifact; the three-dataset evaluation reported here (Section 5) is designed specifically to make that distinction possible.

We propose Essence, a framework that embeds a user’s consumed items into a semantic space using a pre-trained sentence encoder, clusters them into K interest profiles via K -means, and generates recommendations by retrieving items similar to the centroid closest to the user’s recent activity. We refer to this as the *peer-isolation property*: no cross-user interaction graph is consulted at any stage. This is a deliberate architectural constraint, not an oversight: it is the mechanism under test. We describe the resulting objective as depth-optimized: the goal is to retrieve items deep in a user’s long-tail interest space, rather than to maximize aggregate engagement on broadly popular items.

The contributions of this work are:

1. The **Essence architecture**: a personal recommendation framework requiring no collaborative signal and no supervised or cross-user recommender training, operating entirely on a single user’s history.
2. A **recency-based active-cluster-selection** mechanism, evaluated against a static mean-

embedding alternative via ablation (Section 5.6) *and* against three recency-only baselines that do not cluster at all (last-item, uniform-average, exponentially-weighted retrieval) — the comparison against simpler, non-clustering alternatives that the original evaluation did not include.

3. A **domain-dependence finding** across three datasets spanning music, e-commerce, and film (Section 5): Essence’s relative standing among ten evaluated systems is associated with the strength of collaborative/popularity signal in the domain (establishing this as predictive would require additional datasets and an independently-defined domain property), evidenced by effect sizes ranging from Essence’s strongest win (Last.fm-1K, beating Random with $d = +0.301$) to its strongest loss (MovieLens-25M, losing to CF-ItemKNN with $d = -0.550$).
4. **Two hand-implemented, from-scratch multi-interest baselines** (MIND, following Li et al. [1]; ComiRec, following Cen et al. [2]) — neither is available in standard recommendation libraries, and neither original paper’s benchmark numbers are directly comparable under this paper’s evaluation protocol (Section 4), so correctness is verified against each model’s architectural specification rather than against a published number (Section 4 describes the verification methodology and a bug it caught in our own implementation).
5. A **statistical-methodology correction**: paired bootstrap significance testing, Benjamini–Hochberg FDR correction, effect sizes, and BCa robustness cross-checks (Section 5.4), replacing an earlier confidence-interval-overlap methodology, applied uniformly across all three datasets and sub-analyses.

2 Related Work

2.1 Collaborative Filtering

Matrix factorisation methods [5] and neural extensions [4] remain dominant in production recommendation. Their principal limitation is popularity bias: items with sparse interactions are poorly represented, resulting in near-zero recall for long-tail items. CF is optimised for the median user: it performs well on what is broadly consumed and comparatively weakly on what is individually resonant.

2.2 Multi-Interest Retrieval

MIND [1] introduced dynamic routing to extract multiple interest capsules from a user’s interaction history, explicitly addressing the failure of a single user vector to represent diverse tastes. ComiRec [2] extended this with explicit diversity controls at serving time.

Essence shares the multi-interest intuition with MIND and ComiRec, the recognition that a user’s preferences cannot be collapsed into a single vector. However, the similarity ends at the surface. The distinction operates at two levels.

Data level. MIND and ComiRec require population-level interaction graphs. Their dynamic routing is trained on millions of (user, item, interaction) triples across the full user base. Cross-user behavioural data is not optional: it is the mechanism by which interest capsules are learned. Essence requires no data from any other user, ever. The K -means centroids are derived entirely from one user’s own history.

Representation level. Because MIND’s item embeddings are trained on co-occurrence patterns across users, what an item *means* in that space is partly defined by who else liked it. An obscure track heard by very few users is underrepresented or invisible in that embedding space, as its co-occurrence signal is too sparse to be learned reliably. Essence uses a pre-trained sentence

encoder on item metadata only. The embedding captures metadata-derived semantic information about the item, independent of how many people have interacted with it. A singleton item can be embedded as readily as a chart-topping one; the quality of that representation depends on how informative the item’s metadata is, not on its interaction count.

This distinction suggests a hypothesis: Essence should recover long-tail items at higher rates than systems whose embeddings are shaped by co-occurrence sparsity. We test this directly against hand-implemented MIND and ComiRec baselines (Section 4), and the result is more qualified than the hypothesis alone predicts: Essence significantly outperforms both on Amazon Books, but on MovieLens-25M both neural baselines substantially outperform Essence instead, by some of the largest effect sizes observed in this study (Section 5.2). The data-level and representation-level distinctions described above are architectural facts regardless of outcome; whether they translate into a long-tail recovery advantage in practice depends on domain conditions, not on the architectural distinction alone.

Table 1 summarises the architectural distinctions between Essence, MIND, ComiRec, and standard CF.

Table 1: Architectural comparison: Essence vs. prior systems. The primary distinction is not the clustering mechanism, it is the data requirement and what each system is structurally able to score. All four systems are empirically evaluated in this work (Section 5), which reports the domain-dependent long-tail recovery result in full.

Property	CF	MIND	ComiRec	Essence
Cross-user data required	Yes	Yes	Yes	No
Multiple interest profiles	No	Yes	Yes	Yes
Gradient-based training required	No	Yes	Yes	No
Can score zero-interaction items	No	No	No	Yes
Embedding source	Co-occurrence	Co-occurrence	Co-occurrence	Item metadata

2.3 Session-Based and Content-Based Methods

BERT4Rec [3] and GRU4Rec [6] model sequential user behaviour using transformer and recurrent architectures, capturing short-term context effectively. Content-based filtering relies on item feature similarity without interaction logs but typically averages preferences into a single vector, losing intra-user diversity.

Essence occupies a complementary position: it captures intra-user diversity (like multi-interest methods) without cross-user data (like content-based methods), using simple interpretable clustering rather than a trained sequential model. We acknowledge that BERT4Rec offers richer sequential modelling in high-interaction settings; Essence targets privacy-first, low-infrastructure deployment contexts where such models are not viable.

2.4 Reproducibility and Progress in Recommender Systems Research

Ferrari Dacrema et al. [12] showed that a majority of neural recommendation papers at top venues failed to outperform simple, well-tuned non-neural baselines when those baselines were given a fair tuning budget, and that a substantial fraction of claimed improvements did not reproduce under careful re-evaluation. A follow-up analysis [13] extended this finding across a broader set of papers and venues, identifying weak baselines, inconsistent evaluation protocols, and selective reporting as recurring, systemic causes of inflated claims of progress in the field. This paper’s own findings are consistent with, and extend, that line of critique to a different architecture family: content-only, no-cross-user-pooling personalization. Essence’s headline

long-tail advantage survives paired bootstrap significance testing on Amazon Books but is directional and not statistically significant on the smaller Last.fm-1K, and on MovieLens-25M — a domain with strong collaborative signal — Essence loses to collaborative filtering, popularity, and both neural multi-interest baselines by the largest effect sizes observed anywhere in this study (Section 5.2). This is the discipline Ferrari Dacrema et al. call for: a fair, statistically grounded comparison against strong, simple baselines, with mixed results reported alongside favorable ones.

3 The Essence Framework

3.1 Notation

Table 2 defines all mathematical notation used in this section. For each symbol a plain-English equivalent is provided.

Table 2: Notation Legend

Symbol	Plain-English Meaning
H_u	All items user u has consumed (full history)
$i_1, i_2, \dots$	Individual items in the user’s history, in order
$\phi(i)$	Embedding of item i , a 384-dim vector describing its content
E_u	All embeddings stacked into a matrix, one row per item
K	Number of interest clusters (validation-selected per dataset: $K = 10$ Last.fm-1K/Amazon Books)
$c_1, \dots, c_K$	The K cluster centres, one per interest profile
c_k^*	Active cluster centre, closest to the user’s recent activity
$\mu(\cdot)$	Mean (average) of a set of vectors
$\cos(\mathbf{a}, \mathbf{b})$	Cosine similarity: how alike two vectors are
$I \setminus H_u$	All items the user has <i>not</i> yet consumed, the candidate pool
R_u	Final ranked recommendation list for user u
M	Number of recommendations to return (we use $M = 10$)

3.2 Problem Formulation

Given a user’s history H_u (all items they have consumed), the goal is to produce a ranked list $R_u \subseteq I \setminus H_u$ of items they have not yet consumed but are likely to enjoy. Essence accomplishes this using only H_u and a shared embedding function $\phi : I \rightarrow \mathbb{R}^d$. No data from any other user is used at any stage.

3.3 Item Embedding

Each item is described by a short text string encoding its content properties. For music: track name and artist name. For books: title and author name. We convert this text into a 384-dimensional vector using the pre-trained sentence encoder `all-MiniLM-L6-v2` [8, 7]. Items that are semantically similar produce similar vectors; items that are stylistically distant produce different vectors. Critically, this embedding is derived from item metadata text, not from interaction patterns. A niche item can be embedded as readily as a popular one; representation quality depends on the informativeness of the metadata rather than on interaction frequency. All embeddings are computed once offline and cached.

Formally, $\phi(i)$ is the embedding of item i . For user u with history $H_u = \{i_1, \dots, i_n\}$ we compute the embedding matrix $E_u \in \mathbb{R}^{n \times d}$, where each row is the embedding of one item.

3.4 Interest Profile Clustering

We apply K -means clustering to E_u with a per-dataset K (selection procedure below), yielding centroids $\{c_1, \dots, c_K\}$. Each centroid represents a distinct interest profile: the semantic centre of gravity of one coherent facet of the user’s taste. A user who consumes jazz, indie rock, and ambient electronica might, for example, end up with one centroid per genre under $K = 3$. The K centroids together form the user’s interest profile $P_u = \{c_1, \dots, c_K\}$.

This is the core differentiator from average-embedding approaches. The mean of all embeddings collapses all interests into a single blurred point that sits between clusters and represents none of them well. K -means preserves the separation between interest modes, enabling targeted retrieval against each mode independently.

K is selected per dataset on a validation split carved from the training data (the last 20% of each user’s chronological training history; the held-out test set is never touched during selection). K is swept over $\{2, 3, 4, 5, 8, 10, 15, 20, 30, 50\}$ (log-spaced above $K = 8$), and the sweep for a given dataset stops at the first K where either (i) validation Recall@10 improves by less than 2% relative to the previous tested K , or (ii) the mean number of items per cluster (training items per user, divided by K , averaged over users with enough history to cluster) falls below 3 — whichever triggers first. The selected K is the validation-Recall@10-maximizing point among the values tested before the stopping rule fires. This yields $K = 10$ for Last.fm-1K and Amazon Books, and $K = 15$ for MovieLens-25M. One methodological note on how the sweep was conducted: the stopping rule was applied moving forward from $K = 8$ (an earlier, already-reported checkpoint in this project’s revision history) rather than from $K = 2$; applied retroactively to the $K \in \{2, \dots, 5\}$ region, the rule would in fact have triggered earlier for Amazon Books and MovieLens-25M, since validation Recall@10 dips slightly between some of those early, closely-spaced values before climbing again at higher K . We report this plainly as a scope decision — continuing the already-established $K = 8$ checkpoint forward, rather than restarting the rule from $K = 2$ — not a result the stopping rule itself produced.

For users with fewer than K items, Essence falls back to the content baseline (single mean embedding).

3.5 Active Cluster Selection

At inference time, Essence selects the active centroid c_k^* as the cluster centre closest to the mean embedding of the user’s r most recent training-set interactions, ordered chronologically (default $r = 10$):

$$c_k^* = \arg \min_k \|c_k - \mu(\phi(h_{n-r+1}), \dots, \phi(h_n))\|_2 \quad (1)$$

This mechanism assumes recent activity is informative about which interest cluster is currently active. We test this assumption directly via ablation (Section 5.6): replacing the recency-based query with the mean embedding of the user’s entire training history degrades long-tail recall substantially on Last.fm-1K, indicating that recency carries signal beyond what the K -means clustering step alone provides. The ablation is the basis for retaining recency as the default mechanism.

3.6 Recommendation Generation

Given the active centroid c_k^* , Essence retrieves the top- M unseen items by cosine similarity:

$$R_u = \text{top-}M \{i \in I \setminus H_u : \cos(\phi(i), c_k^*)\} \quad (2)$$

This retrieval step operates entirely in embedding space with no collaborative signal. An item’s probability of appearing in R_u is determined solely by its latent similarity to the user’s current interest cluster, not by how many other users have interacted with it.

3.7 Architecture Overview

Figure 1 illustrates the full Essence pipeline from raw user history to final recommendations.

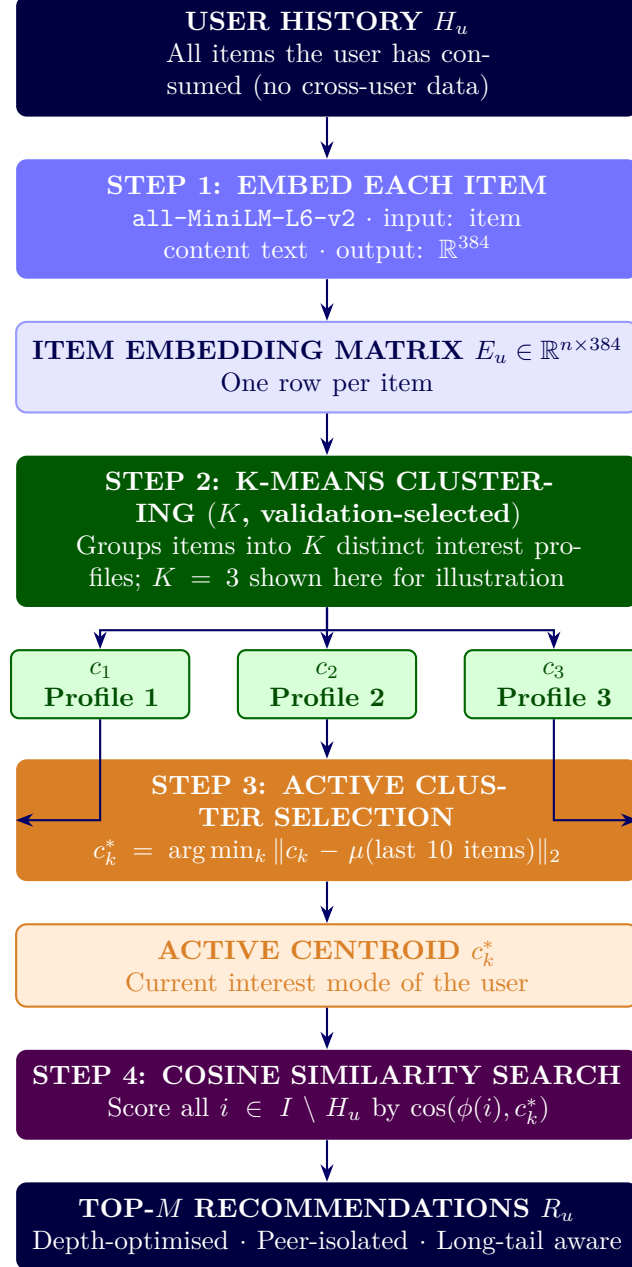


Figure 1: Essence recommendation pipeline. No cross-user data is consulted at any stage (peer-isolation property).

4 Experimental Setup

4.1 Datasets

Last.fm-1K (music). Contains listening histories from 992 users across approximately 19 million events [9]. After filtering to users with 50–500 unique track interactions and removing duplicate (user, track) pairs, our evaluation subset contains 99 users and 22,767 unique tracks — this is the full filtered dataset, not a subsample: no further sampling is applied after the interaction-count filter. A positive interaction is any recorded play; there is no minimum play count or other threshold beyond the 50–500 per-user interaction-count filter. Item embeddings are built from track title + artist name (metadata only; no interaction signal enters the embedding).

Amazon Books (e-commerce). The Amazon Reviews 2023 Books 5-core dataset [10], which guarantees every user and item has at least 5 global interactions. After filtering to users with 20–200 unique item interactions and subsampling to 2,000 users (seed 42), our evaluation subset contains 2,000 users and 61,727 unique items. Item embeddings use title + author name + description, following the same metadata-only approach as Last.fm-1K. A positive interaction is any rating or review in the 5-core file; no minimum star-rating value is required (a 1-star and a 5-star rating both count equally as an interaction). The 20–200-interaction filter and the 2,000-user subsample size are applied identically, with the same seed, to MovieLens-25M below — fixed a priori and held constant across both datasets rather than tuned per dataset, to keep the evaluation (including MIND/ComiRec training and the full bootstrap/FDR suite) computationally tractable at the same scale on both.

MovieLens-25M (film). A 2,000-user subsample (seed 42, same 20–200-interaction filter and sampling convention as Amazon Books) drawn from the MovieLens-25M ratings dataset. Item text is built from title + genres, plus overview text from The Movie Database (TMDb) when available; of 7,724 candidate items identified from the sampled users’ interactions, 7,654 had a TMDb record successfully fetched at all (99.1%), and these form the final catalog — the remaining 70 items, for which the TMDb API lookup itself failed, are excluded from the candidate pool entirely (not merely missing overview text). Within the 7,654 fetched items, any with an empty overview field fall back to title + genres only rather than being dropped. Our evaluation subset contains 2,000 users and 7,654 catalog items. A positive interaction is any rating in the sampled ratings file; no minimum star-rating value is required.

Data attribution. This product uses the TMDb API but is not endorsed or certified by TMDb.

4.2 Train/Test Split

For each user on all three datasets, interactions are sorted chronologically and split per-user: the first 80% form the training set, and the final 20% form the test set. This chronological split is required for internal consistency with the active-cluster-selection mechanism (Section 3.5), which defines “recent” interactions in terms of training-set recency; a random split would make this definition incoherent, since a randomly held-out test item could chronologically precede items retained in training. The split is computed independently per user, with no shared global time cutoff across users: each user’s train/test boundary is defined purely by their own interaction history, not by any dataset-wide date. This preserves within-user chronology but does not enforce a global temporal cutoff; possible cross-user temporal leakage for population-based baselines (CF, Popularity) — an item’s co-occurrence signal can include other users’ interactions that occurred, in wall-clock time, after a given user’s personal train/test cutoff — is therefore a limitation of this evaluation protocol, not eliminated by it (Section 7).

Item embeddings are computed once, offline, over the full item catalog ($\text{train} \cup \text{test}$), so that no test item is excluded from the retrieval candidate pool due to missing embedding coverage.

This is independent of the train/test split decision and applies identically under either splitting strategy.

4.3 Long-Tail Definition

Long-tail items are those with insufficient interaction signal for collaborative filtering to represent reliably. We define long-tail operationally per dataset, anchored to each dataset’s interaction density:

- **Last.fm-1K.** Items with a training-set interaction count of exactly 1 (*singleton tracks*): 16,761 of the 18,679 items appearing in the training set (89.7%). The full candidate pool ($\text{train} \cup \text{test}$) is 22,767 tracks.
- **Amazon Books.** Items with a training-set interaction count of exactly 1 (*singleton items*): 42,878 of the 50,924 items appearing in the training set (84.2%). The full candidate pool ($\text{train} \cup \text{test}$) is 61,727 items. Despite the 5-core guarantee of at least 5 global interactions, subsampling to 2,000 users reduces most items to a single observed interaction in the subset’s training split.
- **MovieLens-25M.** Items with a training-set interaction count of exactly 1: 2,337 of the 6,745 items appearing in the training set (34.6%). Unlike Last.fm-1K and Amazon Books, where singleton items concentrate at the rare-popularity end of the catalog (deciles 3–9), MovieLens-25M’s singletons concentrate more centrally, in deciles 2–5 (32.7% and 32.8% of all singletons falling in deciles 3 and 4 alone) — a dataset-specific difference in decile geometry, not a methodological inconsistency.

LT-Recall@10 is computed only over users with at least one long-tail test item; the remaining users contribute no LT-Recall@10 observation for that metric. Eligible users: 92 of 99 on Last.fm-1K, 1,292 of 2,000 on Amazon Books, 490 of 2,000 on MovieLens-25M.

Generalisation principle. The long-tail threshold should be set at the point where collaborative filtering loses reliable signal, empirically where item interaction count falls below the minimum required for stable co-occurrence estimation. On Last.fm-1K and Amazon Books, the singleton threshold (train interaction count = 1) proved the most meaningful boundary: alternative thresholds either collapsed toward the full catalog (losing discriminative power) or excluded too few items; the same threshold was applied to MovieLens-25M without modification and produced a comparably non-degenerate singleton population (34.6% of training-set items), though alternative thresholds were not separately swept on this third dataset. The key invariant is that long-tail items are defined on the *training set only*, after the train/test split, ensuring no leakage from held-out interactions into the long-tail classification.

LT-Recall@10 is computed only over test items falling in the long-tail set. Users with no long-tail test items are excluded from this metric.

4.4 Systems Compared

- **Random.** Recommends M items drawn uniformly from the unseen candidate pool. Establishes the floor against which all other systems are compared.
- **Popularity.** Recommends the M globally most-interacted-with items the user has not consumed. A non-personalized baseline; included for comparison but not treated as collaborative filtering.
- **CF (ItemKNN).** An item-based collaborative filtering baseline, computed via cosine similarity over the sparse user-item interaction matrix. Recommends items most similar, by

co-occurrence pattern, to items the user has already consumed. No neighbor-count cutoff or shrinkage term is applied: a candidate’s score is the cosine-weighted sum over all training items with any co-occurring user, i.e. full-neighborhood ($k = \infty$) ItemKNN rather than a truncated top- k variant.

- **Content (Average Embedding).** Computes the mean of all the user’s item embeddings and retrieves the M most similar unseen items by cosine similarity. Standard content-based filtering without interest decomposition.
- **Essence (validation-selected K).** Clusters user history into K centroids ($K = 10$ Last.fm-1K/Amazon Books, $K = 15$ MovieLens-25M; selection procedure in Section 3), selects the active centroid by Equation (1), and retrieves top- M unseen items by Equation (2).
- **Last-Item.** Recommends the M unseen items most similar, by cosine similarity, to the single most recent item in the user’s training history. The simplest possible recency-based retrieval: no averaging, no decay, no clustering.
- **Avg-Last-10.** Recommends the M unseen items most similar to the uniform mean embedding of the user’s last 10 training interactions.
- **Recency-Weighted.** Recommends the M unseen items most similar to an exponentially-decay-weighted mean of the user’s last 10 training interactions (decay = 0.9; the most recent item receives weight $\propto 0.9^0$, the tenth-most-recent $\propto 0.9^9$, renormalized to sum to 1). Last-Item, Avg-Last-10, and Recency-Weighted use exactly the same item embeddings as Content and Essence; the only difference across all five is how the query vector is constructed from recent history.
- **MIND.** A hand-implemented dynamic-routing multi-interest network, following Li et al.’s original architecture [1] ($K = 4$ interest capsules, a bilinear routing matrix shared across capsules, 3 routing iterations, routing logits initialized $b \sim \mathcal{N}(0, 0.01^2)$). Neither MIND nor ComiRec is available in standard recommendation libraries, and neither original paper’s reported benchmark numbers are directly comparable here: Li et al. benchmark on an internal industrial dataset and Amazon Books, and Cen et al. on Amazon Books, Taobao, and a mobile-Taobao set, all under sampled-negative evaluation (one positive against roughly 100–1,000 negatives); none report Last.fm-1K at all, and Amazon Books here uses this paper’s own 2,000-user subsample, full-catalog ranking, and no negative sampling (Section 4) — a harder protocol under which lower absolute recall than the original papers’ is expected by construction, not evidence of a broken implementation. We therefore verify correctness against each model’s architectural specification directly, rather than by matching a published benchmark number: a synthetic conformance test for MIND’s routing mechanism, and a directional sanity check (Recall@10 convincingly above Random and Popularity, and within a plausible order of magnitude of the published ranges once the protocol difference is accounted for) applied to both models’ final results. The conformance test caught a symmetry-breaking bug in *our implementation* — not a flaw in the original architecture: an initial version zero-initialized b , which, because the shared bilinear matrix makes the pre-routing representation identical across capsules, guaranteed every capsule updated identically, silently collapsing the model’s effective interest count to 1 regardless of the nominal K . The test (`test_mind_routing.py`) caught this before any number reported here was generated; catching a real, non-obvious bug is the evidence the verification method works, not just a disclosed limitation.
- **ComiRec (SA).** A hand-implemented self-attentive multi-interest network, following Cen et al.’s original architecture [2] ($K = 4$ interest capsules, per-capsule learned query projections implemented as a single K -output linear layer, softmax attention over sequence positions).

ComiRec’s per-capsule projections are independently initialized rather than derived from a shared matrix, so it has no equivalent to MIND’s symmetry-breaking failure mode and was checked with the directional sanity check described above rather than a dedicated conformance test; no comparable bug was found.

- **MIND/ComiRec training.** Both models share the same fixed, untuned training procedure, taken from the original papers’ recommended ranges rather than grid-searched: Adam optimizer, learning rate 10^{-3} , batch size 64, max sequence length 50, up to 50 epochs with early stopping (patience 5) on validation loss over each user’s held-out last training interaction. The training objective is a sampled-softmax loss: the positive item’s score plus 100 uniform-random negative items’ scores (drawn item-uniformly, excluding the padding index) are passed through a cross-entropy loss against the positive. Item and interest-capsule embeddings are 384-dimensional, matching the shared `all-MiniLM-L6-v2` embedding space used by every other system.

Both MIND and ComiRec are trained across 3–4 seeds per dataset to confirm the domain-dependence pattern (Section 5.2) is not seed-specific (Section 5.3).

4.5 Metrics

What we are measuring. Each system produces a ranked list of 10 recommended items per user. We know which items that user actually consumed in the test set (ground truth). The evaluation asks: how much of that ground truth did the system recover?

This is a retrieval problem, not a classification problem. All systems are evaluated against the *full item catalog* with no negative sampling, following the full-ranking evaluation protocol [11]. Absolute recall values are low by construction under this protocol: the meaningful signal is relative comparison across systems evaluated under identical conditions.

- **Recall@10.** Fraction of the user’s test items appearing in the top-10 recommendations.

$$\text{Recall@10} = \frac{|R_u \cap \text{Test}_u|}{|\text{Test}_u|}$$

Example: 20 test items, 2 hits in top-10. Recall@10 = 0.10. Averaged across all users.

- **Long-Tail Recall@10 (LT-Recall@10).** Recall@10 restricted to singleton test items only.

$$\text{LT-Recall@10} = \frac{|R_u \cap \text{LT-Test}_u|}{|\text{LT-Test}_u|}$$

A long-tail hit means the system recommended a niche item that this specific user genuinely consumed, without any popularity signal and without any other user’s behaviour to guide it. This metric directly tests Essence’s core hypothesis.

- **Lift over random.** Recall@10 divided by expected random recall ($M/|I|$). Normalises for candidate pool size and makes cross-dataset comparison interpretable.

4.6 Implementation

All systems implemented in Python 3.11. Embeddings via `sentence-transformers` (`all-MiniLM-L6-v2`, 384 dims). *K*-means via `scikit-learn` (`n_init=10`, `random_state=42`). MIND and ComiRec are implemented from scratch in PyTorch, since neither is available in RecBole or comparable standard recommendation libraries; both are trained per dataset. All ranking steps break score ties deterministically (stable sort by item index), so reported results are exactly

reproducible; this matters for sparse baselines such as CF (ItemKNN), where many candidate items receive identical scores. Significance testing methodology (paired bootstrap, FDR correction, effect sizes, BCa cross-checks) is described once, in Section 5.4, and applied identically across all three datasets rather than being re-derived per dataset. Code available at: <https://github.com/biditdas18/essence>

5 Results

5.1 Headline Results, All Three Datasets

Table 3 and Table 4 report Recall@10 and LT-Recall@10 for all ten systems on all three datasets (99 users on Last.fm-1K, 2,000 users on Amazon Books, 2,000 users on MovieLens-25M; chronological 80/20 split per user throughout; catalog sizes and long-tail definitions in Section 4). Essence’s rank among the ten systems is noted next to its own value in each column.

Table 3: Headline Recall@10, all three datasets ($M = 10$). Best result per dataset in **bold**. Essence’s rank (of 10) is shown in parentheses. Essence uses its validation-selected K per dataset ($K = 10$ Last.fm-1K/Amazon Books, $K = 15$ MovieLens-25M; Section 3).

System	Last.fm-1K	Amazon Books	MovieLens-25M
Random	0.0001	0.0001	0.0012
Popularity	0.0023	0.0021	0.0517
CF (ItemKNN)	0.0018	0.0031	0.0850
Content (Avg Embedding)	0.0038	0.0173	0.0043
Last-Item	0.0265	0.0311	0.0091
Avg-Last-10	0.0252	0.0235	0.0064
Recency-Weighted	0.0304	0.0260	0.0076
MIND	0.0078	0.0052	0.0423
ComiRec (SA)	0.0137	0.0056	0.0706
Essence (val. K)	0.0281 (2nd)	0.0253 (3rd)	0.0053 (8th)

Table 4: Headline LT-Recall@10, all three datasets ($M = 10$). Best result per dataset in **bold**. Essence’s rank (of 10) is shown in parentheses. Essence uses its validation-selected K per dataset, as above.

System	Last.fm-1K	Amazon Books	MovieLens-25M
Random	0.0000	0.0000	0.0000
Popularity	0.0000	0.0000	0.0000
CF (ItemKNN)	0.0000	0.0137	0.0037
Content (Avg Embedding)	0.0029	0.0124	0.0020
Last-Item	0.0064	0.0215	0.0031
Avg-Last-10	0.0134	0.0196	0.0026
Recency-Weighted	0.0151	0.0179	0.0046
MIND	0.0021	0.0057	0.0000
ComiRec (SA)	0.0050	0.0121	0.0014
Essence (val. K)	0.0142 (2nd)	0.0207 (2nd)	0.0030 (4th)

Essence’s standing is not uniform across datasets. On Amazon Books it is 3rd of 10 on Recall@10 and 2nd of 10 on LT-Recall@10 (behind only Last-Item on both metrics; statistically tied with,

not significantly behind, Recency-Weighted on Recall@10 — Section 5.4); on Last.fm-1K it is 2nd of 10 on both Recall@10 and LT-Recall@10, not significantly different from any of the three recency-only baselines (numerically ahead of Last-Item and Avg-Last-10, behind Recency-Weighted); on MovieLens-25M it is 8th of 10 on Recall@10, ahead of only Content and Random, and 4th on LT-Recall@10. Full significance testing for every pairwise comparison in these tables is reported in Section 5.4.

5.2 The Domain-Dependence Pattern

Essence’s rank is not randomly scattered across the three datasets: it is associated with the strength of collaborative and popularity signal available in each domain, measured by how well the non-personalized Popularity baseline and the CF (ItemKNN) baseline perform in that domain (Table 5). Establishing this association as predictive would require additional datasets and an independently-defined domain property, not one measured post hoc via baseline performance, as here.

Table 5: Essence’s rank (of 10 systems, Recall@10) against the strength of collaborative/popularity signal in each domain.

Dataset	Essence rank	CF / Popularity signal
Amazon Books	3rd (beats 7, ties Recency-Weighted, loses to Last-Item)	Weak (CF 0.0031, Popularity 0.0021)
Last.fm-1K	2nd (beats 8, ties Recency-Weighted)	Weak (CF 0.0018, Popularity 0.0023)
MovieLens-25M	8th (beats only Content, Random)	Strong (CF/Pop./ComiRec/MIND are top 4)

Where collaborative and popularity signal is structurally weak, Essence sits in the upper half of the field on both datasets. On Amazon Books it significantly beats CF-ItemKNN, the non-personalized baselines, Content, and both neural multi-interest baselines on Recall@10 (Cohen’s d 0.099–0.282); it is statistically tied with Recency-Weighted (not a significant difference) and significantly behind only Last-Item. On Last.fm-1K, at its validation-selected $K = 10$, Essence significantly beats CF-ItemKNN, the non-personalized baselines, Content, MIND, and ComiRec on Recall@10 (Cohen’s d 0.220–0.301) — at $K = 3$, Essence’s Recall@10 was statistically tied with ComiRec (0.0119 vs. 0.0137); at the validation-selected K , Essence’s own Recall@10 rises to 0.0281 and the tie resolves into a significant win. Essence is not significantly different from any of the three recency-only baselines on Last.fm-1K, including Recency-Weighted, despite trailing it numerically. (LT-Recall@10 shows no significant differences in either direction on Last.fm-1K at this sample size, $n = 99$; see Section 5.8.) Where collaborative signal is strong (MovieLens-25M, driven almost entirely by densely-rated popular titles: item-based CF reaches 0.0850 Recall@10, the single highest value recorded in this study), Essence falls to 8th of 10, losing to CF, Popularity, and both neural baselines by the four largest-magnitude effect sizes observed anywhere in this evaluation (Cohen’s d from -0.378 to -0.550 , all four significant after BH-FDR correction; every other significant effect size across Last.fm-1K and Amazon Books stays at or under $|d| \approx 0.30$).

This pattern is consistent and well-evidenced across all three datasets. We treat it as a genuine domain-dependence finding rather than an architectural strength: it is Essence’s *relative standing* that tracks domain conditions, not necessarily anything specific to the K -means clustering mechanism as opposed to the underlying content embeddings more generally (Section 5.8

addresses this directly).

5.3 MIND/ComiRec Multi-Seed Robustness

Table 6 reports MIND and ComiRec Recall@10 across independent training seeds per dataset (the canonical seed 42 plus 2–3 additional seeds), to check whether the domain-dependence pattern above depends on a single training run.

Table 6: MIND/ComiRec Recall@10 across training seeds. “Beats Essence” counts seeds where the baseline’s Recall@10 exceeds Essence’s own Recall@10 for that dataset (Table 3).

Dataset (Essence Recall@10)	System	Mean $\pm$ Std	n	Beats Essence
Last.fm-1K (0.0281)	MIND	0.0067 ± 0.0014	4	0/4
	ComiRec (SA)	0.0116 ± 0.0020	4	0/4
Amazon Books (0.0253)	MIND	0.0053 ± 0.0004	4	0/4
	ComiRec (SA)	0.0064 ± 0.0007	4	0/4
MovieLens-25M (0.0053)	MIND	0.0464 ± 0.0036	3	3/3
	ComiRec (SA)	0.0607 ± 0.0086	3	3/3

Essence’s Recall@10 in this table uses its validation-selected K per dataset (Table 3), not the $K = 3$ value used in an earlier revision of this analysis; at $K = 3$, ComiRec (SA) beat Essence on 2 of 4 Last.fm-1K seeds, which this paper originally characterized as a statistical tie. At the validation-selected $K = 10$, Essence’s own Recall@10 rises enough ($0.0119 \rightarrow 0.0281$) that ComiRec no longer beats it on any seed. The domain-dependence pattern itself is not an artifact of a single seed: MIND never outranks Essence on Last.fm-1K or Amazon Books and always outranks Essence on MovieLens-25M, across every seed tested. ComiRec (SA) now shows the same clean pattern on all three datasets (never on Last.fm-1K or Amazon Books, always on MovieLens-25M) rather than being seed-dependent anywhere.

5.4 Statistical Validation Methodology

Every comparison reported in this paper uses a paired bootstrap significance test on the per-user difference between two systems (10,000 resamples over users, seed 42; observed mean difference, 95% percentile CI, and a paired Cohen’s d computed as $\text{mean}(\text{difference})/\text{std}(\text{difference})$). Independent per-system confidence intervals are not used: they do not test the quantity of interest directly (whether one system’s per-user metric exceeds another’s) and can disagree with the paired test on borderline cases. Where multiple related comparisons are tested together, we apply Benjamini–Hochberg (BH) false-discovery-rate correction at $q < 0.05$ within that family, and cross-check every raw-significant result under bias-corrected and accelerated (BCa) bootstrap resampling to confirm the conclusion is not an artifact of the simpler percentile method.

Correction families. The Last.fm-1K/Amazon Books baseline comparison (Essence vs. each of the nine other systems, per dataset, per metric; 36 comparisons) and the multimodality-stratification test (Essence vs. Recency-Weighted, stratified by each user’s own per-history silhouette score into low/medium/high clustering-quality tertiles, per dataset, per metric; up to 18 comparisons, Section 5.8) are corrected together as a single Benjamini–Hochberg family (54 comparisons nominally; 52–53 in practice, see below): pooling them changes no comparison’s significance verdict relative to correcting them separately, so we report the simpler pooled version. The MovieLens-25M baseline comparison (18 comparisons, added to this evaluation after the 36 above) and the decile-level cold-start analysis (Section 5.5; Essence, and separately each recency-only baseline, vs. CF-ItemKNN, restricted to the rarest-item deciles) are each corrected as their

own family: the cold-start analysis asks a narrower, mechanistically distinct question, isolating a single structural property of CF (zero co-occurrence signal for zero-training-interaction items) rather than aggregate performance. One correction to the stratification family’s comparison count: one or two of the up-to-18 stratum/metric cells are occasionally degenerate — every user in that cell scores exactly zero on *both* Essence and Recency-Weighted (most often the low- or medium-silhouette stratum on the smaller LT-Recall@10 subsets), giving zero variance in the paired difference. The standard two-sided bootstrap p -value formula is undefined in this case and, applied naively, returns $p = 0$ (spuriously “significant”); these cells are excluded from the pooled family rather than included as evidence of a real difference, since there is no difference to detect. This yields 53 comparisons at $K = 3$ and 52 at the validation-selected K (Section 3), not 54 in either case.

5.5 Cold-Start vs. Singleton Long-Tail: A Scope Boundary

LT-Recall@10, as defined in Section 4, treats every training-set singleton as long-tail without distinguishing *how* singleton an item is. A finer-grained popularity-decile breakdown reveals that this hides two very different regimes.

Table 7: Essence vs. CF-ItemKNN at the two rarest popularity deciles, all three datasets, at Essence’s validation-selected K per dataset. Positive diff favors Essence. Paired bootstrap, 10,000 resamples; BH-FDR within this 6-comparison family.

Dataset	Decile	Essence	CF	Diff	Cohen’s d	Sig. (FDR)
Last.fm-1K ($K = 10$)	1	0.0221	0.0000	+0.0221	0.246	Yes
Last.fm-1K ($K = 10$)	2	0.0352	0.0000	+0.0352	0.351	Yes
Amazon Books ($K = 10$)	1	0.0215	0.0000	+0.0215	0.228	Yes
Amazon Books ($K = 10$)	2	0.0224	0.0004	+0.0220	0.193	Yes
MovieLens-25M ($K = 15$)	1	0.0092	0.0000	+0.0092	0.108	Yes
MovieLens-25M ($K = 15$)	2	0.0029	0.0047	−0.0019	−0.026	No

At true cold-start (decile 1: items with *zero* training-set interactions, not merely rare ones), Essence significantly beats CF-ItemKNN on all three datasets, and this pattern is unchanged from an earlier fixed- $K = 3$ version of this analysis (in which MovieLens-25M’s decile-1 result was a marginal pass rather than a clean one; at the validation-selected $K = 15$ it is unambiguous, $q = 0.002$). But this is not evidence for Essence’s architecture specifically: re-running the identical test with Recency-Weighted, Last-Item, or Avg-Last-10 in Essence’s place, 15 of 18 comparisons are also significant, including every Last.fm-1K and Amazon Books comparison for all three non-Essence baselines (this check does not depend on Essence’s K and was not re-run). The mechanism is mechanical, not architectural: CF-ItemKNN’s co-occurrence score is *exactly* 0.0000 for any item with zero training interactions by construction, so any method that scores items via content embeddings — clustering or not — clears that bar trivially. The defensible claim is therefore narrower than “Essence solves cold-start”: Essence, like every content-embedding-based method tested, beats collaborative filtering at true cold-start.

Separately, on Amazon Books, an earlier fixed- $K = 3$ version of this analysis found Essence losing to Last-Item at both deciles 1 and 2 (significant after FDR, $d = -0.093$ and -0.065) and to Recency-Weighted at decile 2 ($d = -0.075$). At the validation-selected $K = 10$, none of these comparisons remain significant: Essence vs. Last-Item is $d = -0.064$ (decile 1, $q = 0.082$) and $d = -0.022$ (decile 2, $q = 0.549$); vs. Recency-Weighted, $d = -0.003$ (decile 1, $q = 0.927$) and $d = -0.023$ (decile 2, $q = 0.549$); vs. Avg-Last-10, neither decile is significant at either K . Essence’s cold-start weakness relative to the recency baselines specifically is therefore weaker

evidence at the validation-selected K than it was at $K = 3$, though the direction (a numerical, non-significant disadvantage) is unchanged. The distributional claim that Essence’s actual LT-Recall@10 hits on Amazon Books concentrate in deciles 3–9 has not been re-verified at $K = 10$ and is not repeated here pending that check. The decile-1/2 cold-start weakness and the singleton-recall headline win remain about largely disjoint item populations at the new K (Table 7 vs. Table 4): not a contradiction, a scope boundary. The long-tail recovery result in Table 4 should be read as covering moderately-rare-but-observed items, not true cold-start items.

5.6 Active Cluster Selection Ablation

Section 3.5 motivates recency-based active cluster selection on the grounds that it captures session-aware context without requiring a trained sequential model. We test this directly by comparing it against an alternative: selecting the active centroid using the mean embedding of the user’s entire training history, rather than the most recent $r = 10$ interactions.

Table 8: Active cluster selection ablation, all three datasets at their validation-selected K (chronological split, $M = 10$).

Dataset	Variant	Recall@10	LT-Recall@10	LT / C
Last.fm-1K ($K = 10$)	Essence (recency, $r = 10$)	0.0281	0.0142	4.84
	Essence (mean-select, full history)	0.0016	0.0005	0.16
	Content (Avg Embedding, reference)	0.0038	0.0029	1.0
Amazon Books ($K = 10$)	Essence (recency, $r = 10$)	0.0253	0.0207	1.67
	Essence (mean-select, full history)	0.0195	0.0118	0.95
	Content (Avg Embedding, reference)	0.0173	0.0124	1.0
MovieLens-25M ($K = 15$)	Essence (recency, $r = 10$)	0.0053	0.0030	1.46
	Essence (mean-select, full history)	0.0044	0.0037	1.83
	Content (Avg Embedding, reference)	0.0043	0.0020	1.0

On Last.fm-1K, mean-select performs below even the Content baseline on long-tail recall (LT/Content $4.84\times$ in recency-select’s favor). This is consistent with the mechanism collapsing toward Content: selecting the centroid nearest a user’s global mean embedding is functionally similar to querying with that mean directly, which is what Content already does, while incurring the additional cost of clustering without a corresponding benefit. On Amazon Books, the same pattern holds but is less extreme: recency-select still leads on every column, but mean-select stays above the Content baseline ($0.95\times$ – $1.67\times$) rather than collapsing below it. On MovieLens-25M, the picture is mixed rather than one-sided: recency-select leads on raw Recall@10 (0.0053 vs. 0.0044), but mean-select is *ahead* on LT-Recall@10 (0.0037 vs. 0.0030 , $1.83\times$ vs. $1.46\times$ Content). No significance testing has been run on these differences; they are reported as raw comparisons, consistent with how this ablation was originally reported for Last.fm-1K alone. Recency-based selection is retained as Essence’s default mechanism on the strength of the Last.fm-1K and Amazon Books results; MovieLens-25M’s mixed result is reported rather than smoothed over, and is consistent with this paper’s broader finding that Essence’s mechanisms behave differently in the strong-collaborative-signal regime that MovieLens-25M represents (Section 5.2).

5.7 Effect of User-Authored Review Embeddings (Amazon Books)

As a secondary experiment, we tested whether building user taste profiles from review text (first-person language describing a user’s reaction to consumed items) rather than item metadata

changes Essence’s relative performance. Review coverage was complete for the 2,000-user Amazon Books subset.

Table 9: Amazon Books: metadata embeddings (Pass 1) vs. user-review embeddings (Pass 2). Pass 2 builds taste profiles from the user’s own review text; candidates are scored using metadata embeddings. This secondary experiment uses $K = 3$ throughout and has not been re-run at Essence’s validation-selected $K = 10$ (Section 3); it is reported here as a fixed- K diagnostic on the embedding-input question, not as a headline result.

Pass	System	Recall@10	LT-Recall@10	LT / Content
Pass 1 (metadata)	CF (ItemKNN)	0.0031	0.0137	—
	Content (Avg)	0.0173	0.0124	1.0×
	Essence ($K = 3$)	0.0221	0.0197	1.59×
Pass 2 (user review)	CF (ItemKNN)	0.0031	0.0137	—
	Content (Avg)	0.0027	0.0012	1.0×
	Essence ($K = 3$)	0.0041	0.0034	2.83×

Absolute recall degrades substantially in Pass 2. Two factors likely contribute. First, an input-distribution mismatch: taste profiles in Pass 2 are built from review-text embeddings, while candidate items are still scored using metadata embeddings. Although both are produced by the same encoder, review text and item metadata occupy different regions of that shared embedding space, and comparing across these regions is less discriminative than comparing inputs of the same type. Second, review text is a noisier input than structured metadata: a product review may describe shipping conditions or personal circumstances unrelated to the item’s content, and an embedding built from off-topic text does not faithfully represent either the item or the user’s response to it.

Essence’s relative advantage over Content persists in Pass 2 (2.83× vs. 1.59× in Pass 1, both above parity), suggesting the K -means clustering mechanism extracts usable interest structure even under degraded input quality. The absolute recall values in Pass 2 are low enough that this comparison carries limited statistical weight; we treat it as a directional observation motivating the embedding-input-curation discussion in Section 7.

5.8 Limitations, Stated as Findings

Three specific, quantified properties of Essence’s behavior are reported here as findings.

The artist/author shortcut. On both datasets where this was tested, at $K = 3$ (not yet re-verified at the validation-selected K), roughly a fifth of Essence’s top-10 recommendations shared an artist (Last.fm-1K, 20.6%) or author (Amazon Books, 19.8%) with an item in the active cluster — not the majority of recommendations, but a disproportionate share of the hits: excluding those same-artist/author candidates collapsed Recall@10 from 0.0119 to 0.0014 on Last.fm-1K and from 0.0221 to 0.0073 on Amazon Books (LT-Recall@10 from 0.0059 to 0.0005 and from 0.0197 to 0.0080, respectively), at $K = 3$. This analysis has not been re-run at Essence’s validation-selected K ($K = 10$ for both datasets; Section 3); we report the $K = 3$ figures as the most recent verified numbers rather than assume they are unchanged. Some of Essence’s apparent long-tail advantage is therefore closer to identity-based retrieval (more of what a known creator made) than to a genuinely novel content match, at least at $K = 3$, though the majority of recommendations are not same-artist/author.

Last.fm-1K’s sample-size fragility. An earlier, small ($K \in \{2, \dots, 5\}$) manual sweep on this dataset found Recall@10 rising sharply between $K = 3$ (0.0119) and $K = 4/K = 5$

(0.0239/0.0251), with K -means seed variance at fixed $K = 3$ (0.0122 ± 0.0011 across 10 seeds) far too small to explain that jump. This was an early signal that $K = 3$ was not a good fixed choice for this dataset specifically — one of the observations that motivated moving to the validation-selection procedure in Section 3 rather than a single fixed K across all three datasets. That procedure now supersedes the manual sweep, selecting $K = 10$ for Last.fm-1K on a held-out validation split. The underlying caveat about sample size is independent of K and still holds: long-tail comparisons throughout Last.fm-1K rest on very few eligible hits (as few as 2 of 92 users per method for the Essence-vs-Content comparison in Table 4), so Last.fm-1K’s LT-Recall@10 comparisons should be read as directional rather than conclusive at any K . Last.fm-1K’s headline numbers should be read with this fragility in mind.

The unresolved clustering-vs-recency-weighted question. The abstract raises, and does not fully resolve, how much of Essence’s advantage where it exists comes from K -means clustering specifically versus recency-weighted retrieval alone. Two results bear on this directly, and both weaken — without reversing — at Essence’s validation-selected K compared to an earlier, fixed- $K = 3$ version of this analysis. First, Essence’s standing against Recency-Weighted on Recall@10 is no longer a clean loss on all three datasets: at $K = 3$, it lost significantly on all three; at the validation-selected K , it loses significantly only on MovieLens-25M ($d = -0.058$), and is statistically tied with (not significantly behind) Recency-Weighted on both Last.fm-1K ($d = -0.127$) and Amazon Books ($d = -0.009$) — the Amazon Books loss was significant at $K = 3$ ($d = -0.059$, $q = 0.021$) and is not at $K = 10$ ($q = 0.779$). Second, the same silhouette-stratification test as before — comparing Essence against Recency-Weighted within low/medium/high per-user clustering-quality tertiles, on the hypothesis that clustering should help most for genuinely multi-modal (high-silhouette) users — is rerun at the new K within the same pooled-family methodology (BH-FDR within the 52-comparison family combining the Last.fm-1K/Amazon Books baseline comparisons and the all-three-dataset stratification test, degenerate zero-variance comparisons excluded; Section 5.4). At $K = 3$, six of the seventeen clean Last.fm-1K/Amazon Books/MovieLens-25M stratification comparisons were significant losses for Essence, including the high-silhouette stratum on Last.fm-1K ($d = -0.315$, Recall@10) and MovieLens-25M ($d = -0.096$, Recall@10). At the validation-selected K (sixteen clean comparisons), only one remains significant: MovieLens-25M’s high-silhouette stratum on LT-Recall@10 ($d = -0.092$). The Last.fm-1K high-silhouette loss and the MovieLens-25M high-silhouette Recall@10 loss are no longer significant ($d = -0.228$ and $d = -0.079$ respectively). The core qualitative conclusion is unchanged — no stratum, on any dataset or metric, at either K , shows Essence significantly *ahead* of Recency-Weighted — but the evidence that clustering *actively hurts* specific identifiable user groups is now substantially narrower than the $K = 3$ version of this analysis suggested: one significant loss, not five. If clustering were adding value specifically by separating genuinely distinct interests, the users most likely to show that value are still exactly the ones for whom it does not appear; the finding is that clustering does not help, more so than that it actively hurts. This does not mean clustering contributes nothing — the active cluster selection ablation (Section 5.6) shows recency-aware selection matters — but it does mean the clustering step itself, isolated from recency weighting, is not empirically supported as the source of Essence’s advantage where an advantage exists.

A separate, small-scale pilot (200 users, MovieLens-25M) testing whether adaptive per-channel weighting between two embedding channels could improve on Recency-Weighted found no directional win and remains unvalidated exploratory work, not reported here as a result.

6 Discussion

6.1 How the Apparent Conclusion Shifted Across the Evaluation

The headline conclusion this paper would have supported changed at every stage additional rigor or scope was added. Table 10 reconstructs this using only numbers already reported above (headline Recall@10, Table 3; significance results, Section 5.2; stratification results, Section 5.8) — no new comparisons are run for this table.

Table 10: How the apparent headline conclusion shifts as evaluation stages are added, using only already-reported numbers. Each row’s evidence is a subset of the final evaluation, not a separate experiment.

Stage	Evidence at this stage	Apparent conclusion at this stage
1. Naive baselines only	Amazon: Essence 0.0253 vs. CF 0.0031, Popularity 0.0021, Content 0.0173, Random 0.0001	Essence clearly beats collaborative filtering and popularity.
2. + recency baselines	Amazon: Last-Item 0.0311 and Recency-Weighted 0.0260 exceed Essence’s 0.0253; Avg-Last-10 (0.0235) does not	Essence’s apparent edge over CF was partly just recency, not clustering — plain recency retrieval beats Essence on 2 of 3 comparisons.
3. + paired bootstrap / FDR	Essence loses <i>significantly</i> to Last-Item ($d = -0.058$) on Amazon but is statistically tied with Recency-Weighted ($d = -0.009$, not significant); still significantly beats CF, Popularity, Content, and both neural baselines	Essence’s real, statistically-supported value is beating collaborative/popularity/neural baselines, and it is not clearly behind simple recency-based retrieval either — only Last-Item remains a significant loss.
4. + third dataset (MovieLens)	Essence ranks 8th/10 on MovieLens (0.0053), behind CF (0.0850), Popularity (0.0517), ComiRec (0.0706), and MIND (0.0423), by effect sizes down to $d = -0.550$	Whether Essence’s Amazon-style advantage holds is associated with how strong collaborative/popularity signal is in the domain, not shown to be predictive of it — not a fixed property of the architecture either way (Section 5.2).
5. + clustering-mechanism stratification	Essence never significantly beats Recency-Weighted in <i>any</i> silhouette stratum on any dataset; of sixteen clean stratum/metric comparisons, only one is a significant loss (MovieLens-25M’s highest-silhouette stratum, LT-Recall@10, $d = -0.092$)	Even where Essence’s relative standing is favorable, the K -means clustering step itself — isolated from recency-weighted retrieval — is not empirically supported as the source of that advantage; the evidence that it actively <i>hurts</i> specific user groups is narrow, not broad.
6. + validation-selected K (replacing a fixed $K = 3$)	Essence’s own numbers rise on Last.fm-1K/Amazon Books at $K = 10$ (Amazon: 0.0221 $\rightarrow$ 0.0253; Last.fm-1K: 0.0119 $\rightarrow$ 0.0281), moving Essence from 5th/4th to 2nd/3rd of 10 on Recall@10; the Amazon Recency-Weighted loss ($d = -0.059$, significant at $K = 3$) becomes a tie ($d = -0.009$); the stratification family’s significant-loss count drops from six of seventeen (at $K = 3$) to one of sixteen (at the new K)	Fixing $K = 3$ without validating it was itself understating Essence’s standing on two of three datasets, and overstating how often clustering “actively hurts” specific user groups. Properly selecting K moves the story further toward domain-dependence and away from a uniform weakness relative to recency retrieval — MovieLens-25M’s result, where CF signal is strong, is essentially unchanged.

Each stage did not overturn the previous one so much as narrow what was actually being claimed: from “clustering beats collaborative filtering” to “recency-aware retrieval mostly matches or beats collaborative filtering, clustering’s own contribution is domain-dependent, and neither the

case for clustering nor the case against it survives an untuned K without narrowing further.” This progression is the argument, developed in this section and Section 5.8, for reporting a mixed, domain-dependent result rather than the Stage-1 comparison alone.

6.2 A Plausible Mechanism: Why Popularity-Driven Discovery May Self-Reinforce

Collaborative filtering’s reliance on cross-user co-occurrence is consistent with a well-documented feedback dynamic in the literature, not directly measured by this paper’s static evaluation: items that already have more interactions can generate more training signal, potentially improving their representation in the model, increasing their likelihood of being recommended, and generating further interactions. Items with few interactions would not benefit from this loop, regardless of how well they might match any individual user’s taste, because the signal needed to learn that match does not exist in sufficient quantity. This paper’s evidence is consistent with, though not direct confirmation of, that dynamic: the non-personalized popularity baseline, which ranks purely by interaction count, records exactly zero long-tail recall on all three datasets (Table 4) — a structural consequence of the baseline’s definition rather than an observation of the loop unfolding over time. Item-based CF records zero long-tail recall on the sparsest dataset, Last.fm-1K, under deterministic ranking; on the denser Amazon Books it surfaces some low-interaction items but pays a large cost to overall recall (0.0031, near the bottom of the field). Essence’s retrieval step does not depend on this loop, since item representations are derived from content rather than interaction frequency: an item’s likelihood of being recommended is a function of embedding similarity to the active cluster, independent of how many other users have interacted with it. MovieLens-25M is the case where this feedback dynamic runs at full strength in the other direction: CF’s Recall@10 (0.0850) is the highest value recorded anywhere in this study, driven almost entirely by a small set of densely-rated popular titles, and Popularity itself ranks 3rd of 10 systems overall (0.0517) — yet even here, CF’s LT-Recall@10 (0.0037) is far below its own overall Recall@10, confirming that strong aggregate performance from co-occurrence signal does not translate into long-tail recovery; it is domain conditions, not a fixed property of collaborative filtering, that determine how strong this feedback dynamic runs (Section 5.2).

6.3 Deployment Positioning: A Candidate-Generation Method, Not a Ranking Replacement — and Not a Universal One

The absolute recall values reported in Section 5 (approximately 0.005–0.03 under full-catalog evaluation) should be interpreted in light of how such a system would actually be deployed. Essence is not proposed as a replacement for a primary, CF-driven recommendation feed; CF baselines remain effective at surfacing broadly popular content efficiently, which is appropriate for a general homepage or default feed. On Amazon Books and Last.fm-1K, Essence is better understood as a specialized retrieval path: a candidate-generation method suited to a dedicated discovery surface (for example, a “more like this” or deep-catalog-exploration interface) where the explicit goal is surfacing items dissimilar from what population-level signals would otherwise recommend.

This positioning does not extend cleanly to every domain, and MovieLens-25M is the counterexample: where collaborative and popularity signal is strong, Essence trails not only CF and Popularity but both neural multi-interest baselines as well, by the largest effect sizes observed in this study (Section 5.2). A deployment decision should be conditioned on the domain’s collaborative signal strength: whether Essence is a good candidate-generation method tracks with measurable properties of the domain, across the three domains evaluated here, rather than being a fixed property of peer-isolated, content-only personalization.

6.4 Privacy and Deployment Constraints

Because Essence consults no cross-user data at any stage, it is deployable in settings where collaborative filtering is not viable: privacy-regulated environments, on-device recommendation without server-side interaction logs, federated settings where data cannot leave the user’s device, and cold-start contexts where a population-level interaction graph does not yet exist. This is a direct consequence of the architecture (Section 3) rather than an added feature, and follows from the same data-level distinction discussed in Section 2.2.

7 Limitations and Future Work

- **Scale.** Last.fm-1K evaluation uses 99 users. Amazon Books and MovieLens-25M each replicate on 2,000 users, but larger-scale evaluation across all three datasets would strengthen generalisability claims. In particular, the Last.fm-1K long-tail comparisons rest on few eligible long-tail hits (as few as 2 of 92 users per method) and are directional rather than statistically substantive (Section 5.8); the Amazon Books and MovieLens-25M results, at 2,000 users each, carry the statistical weight.
- **Cross-user temporal leakage.** The chronological split (Section 4) is computed independently per user, with no shared global time cutoff. This preserves within-user chronology but does not enforce a global temporal cutoff; possible cross-user temporal leakage for population-based baselines (CF, Popularity) — an item’s co-occurrence signal can include other users’ interactions that occurred, in wall-clock time, after a given user’s personal train/test cutoff — is therefore a limitation of this evaluation protocol, not eliminated by it.
- **K -selection anchoring.** K is now validation-selected per dataset (Section 3: $K = 10$ Last.fm-1K/Amazon Books, $K = 15$ MovieLens-25M), replacing a fixed $K = 3$. The one real caveat on the selection procedure itself: the stopping rule was applied moving forward from an already-established $K = 8$ checkpoint, not from $K = 2$. Applied retroactively to the $K \in \{2, \dots, 5\}$ region, the rule would in fact have triggered earlier for Amazon Books and MovieLens-25M, since validation Recall@10 dips slightly across some of those closely-spaced early values before climbing again at higher K ; this is disclosed as a scope decision in continuing an existing checkpoint forward, not a result the stopping rule itself produced from a clean start.
- **Active cluster selection.** The recency-based heuristic is simple by design. A learned selection mechanism could improve session-aware accuracy without requiring cross-user training.
- **Filter bubble risk.** Peer-isolation removes the serendipity that cross-user signals can provide. Whether repeated centroid queries cause semantic narrowing is an open empirical question.
- **Embedding input curation.** Section 5.7 demonstrates that noisy embedding inputs degrade performance. The quality of what enters the embedding determines the quality of what Essence can find. Automated curation of embedding input text (filtering off-topic or irrelevant content before encoding) is a direct practical extension.
- **Audio-based embeddings.** Current embeddings derive from item metadata text only. For music, incorporating audio-based representations (capturing sonic texture, timbre, rhythm, and harmonic structure directly from audio signal) would more precisely model the latent acoustic fingerprint underlying individual taste. This represents the primary planned extension of Essence: replacing or augmenting text embeddings with audio embeddings and measuring

whether long-tail recall improves when the embedding captures what the music actually sounds like rather than what it is called.

- **Cross-space embedding alignment.** Pass 2 results suggest that aligning review-text and metadata input distributions within the shared embedding space (via contrastive training or a learned projection layer) could unlock the full signal of user-authored representations while maintaining retrieval coherence.
- **Hand-implemented, not pre-trained, multi-interest baselines.** MIND and ComiRec are evaluated here (Section 4) via from-scratch PyTorch implementations trained per dataset, since neither is available in standard recommendation libraries, rather than via pre-trained, production-tuned checkpoints. A well-tuned, larger-scale MIND or ComiRec deployment could plausibly outperform the versions evaluated here, particularly on MovieLens-25M, where both already outrank Essence substantially.
- **Rating threshold.** All Amazon and MovieLens ratings/reviews are treated as positive interactions regardless of rating value; this evaluates engagement/consumption prediction rather than preference prediction specifically. A sensitivity analysis using a positive-rating threshold (e.g., rating ≥ 4) was not performed due to time constraints and is left to future work.
- **The domain-dependence mechanism is not fully explained.** “Collaborative/popularity signal strength” (Section 5.2) is measured post hoc via the baselines’ own performance in each domain, not from an independent domain property decided in advance. A more principled, dataset-independent characterization of when peer-isolated personalization should be expected to help is left to future work.

8 Conclusion

We presented Essence, a personal embedding cluster framework that generates recommendations by clustering a user’s item history into K semantic centroids and retrieving items by similarity to the centroid closest to the user’s recent activity, without consulting any cross-user data. We evaluated Essence on three datasets spanning music, e-commerce, and film (Last.fm-1K, Amazon Books, MovieLens-25M) against nine baselines: a non-personalized popularity baseline, a real collaborative-filtering baseline, a content-based average-embedding baseline, three recency-based baselines that use the same embeddings as Essence but no clustering, and two hand-implemented multi-interest neural baselines (MIND, ComiRec).

Essence’s performance relative to these baselines is domain-dependent: across the three evaluated datasets, Essence’s rank among the ten evaluated systems is associated with the strength of collaborative/popularity signal available in the domain (Section 5.2); establishing this as predictive would require additional datasets and an independently-defined domain property, not one measured post hoc via baseline performance. On Amazon Books and Last.fm-1K, where that signal is structurally weak, Essence sits in the upper half of the field, significantly beating collaborative filtering, popularity, content-averaging, and both neural baselines on Recall@10 or LT-Recall@10 in at least one dataset, and is statistically tied with (not significantly behind) most of the recency-based retrieval baselines that use the identical embeddings without any clustering step — all three on Last.fm-1K, two of three on Amazon Books, where only Last-Item remains a significant loss. On MovieLens-25M, where collaborative signal is strong, Essence ranks 8th of 10 systems, losing to collaborative filtering, popularity, and both neural baselines by the largest effect sizes observed anywhere in this study.

Whether peer-isolated, content-only personalization helps is associated with how strong collabo-

rative and popularity signal already is in a given domain, across the three domains evaluated here; this association is not established as predictive of behavior on domains not yet tested. A targeted stratification test (Section 5.8) finds no evidence that the K -means clustering step, isolated from recency-weighted retrieval, is itself the source of Essence’s advantage where an advantage exists. Essence is a data point on when peer-isolated, content-only personalization is and isn’t the right architectural choice, not a universally superior candidate generator.

An ablation (Section 5.6), now run on all three datasets at their validation-selected K , indicates that Essence’s long-tail advantage, where present, depends at least partly on its recency-based active cluster selection mechanism specifically, not on K -means clustering alone: replacing recency-based selection with a static mean-embedding query degrades performance below the content baseline on Last.fm-1K and stays close to the content baseline (rather than clearly above it) on Amazon Books. On MovieLens-25M the result is mixed — recency-select leads on Recall@10 but mean-select leads on LT-Recall@10 — so this mechanism’s benefit is not uniform across domains.

The long-tail definition used here (items with exactly one training-set interaction) was applied independently, without modification to the threshold or its derivation, to three datasets spanning different domains, scales, and interaction densities. The resulting singleton populations differ substantially in where they sit within the catalog — concentrated at the rare-popularity end on Last.fm-1K and Amazon Books (deciles 3–9), but more centrally on MovieLens-25M (deciles 2–5, Section 4) — a dataset-specific difference in decile geometry that the cold-start analysis (Section 5.5) shows matters for interpreting long-tail results correctly, not a failure of the definition itself.

Limitations of the present evaluation, beyond those above, are discussed in Section 7, including dataset scale, sensitivity to the choice of K , and the increased recall degradation observed when taste profiles are built from noisy, user-authored text (Section 5.7) rather than structured item metadata.

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